
Factors Associated with Inadequate Sexual and Reproductive Health Education Among Adolescents and Its Consequences in Selected Secondary Schools in Douala, Cameroon

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Received: May 20, 2026

Accepted: May 27, 2026

Online Published: Jun 03, 2026

Abstract

Sexual and reproductive health (SRH) education remains a major public health concern among adolescents, particularly in developing countries where socio-cultural barriers, inadequate school-based programs, and poor parent–child communication limit access to accurate reproductive health information. This study assessed factors associated with inadequate sexual and reproductive health education and its consequences among adolescents in selected secondary schools in Douala, Cameroon. A descriptive cross-sectional study was conducted among 378 adolescents recruited from three secondary schools using a non-probability sampling technique. Data were collected using a structured and pretested questionnaire and analyzed using descriptive and inferential statistics. Chi-square tests were performed to determine associations between selected variables, with statistical significance set at $p < 0.05$. The majority of respondents were female (74.0%) and aged 15–16 years (48.2%). Major factors associated with inadequate SRH education included the absence of comprehensive school-based sexuality education programs (57.8%), poor parent–child communication regarding sexuality (56.6%), inadequate financial resources (47.4%), and media/social media influence (36.3%). Inferential analysis revealed that lack of school-based SRH programs ($\chi^2 = 22.43$; $p < 0.001$), poor parent–child communication ($\chi^2 = 18.76$; $p < 0.001$), media influence ($\chi^2 = 7.15$; $p = 0.008$), and cultural restrictions ($\chi^2 = 6.37$; $p = 0.012$) were significantly associated with inadequate SRH education. The most frequently reported consequences were sexually transmitted infections (62.5%) and early or unintended pregnancies (53.4%), followed by school dropout (38.0%), unsafe abortions (29.0%), and sexual exploitation (26.0%). Most respondents expressed the need for counseling, classroom discussions, and guidance from parents, teachers, and health professionals. Strengthening comprehensive sexuality education and improving parent–adolescent communication may significantly improve adolescent sexual and reproductive health outcomes in Cameroon.

Keywords: Adolescents, sexual and reproductive health, sexuality education, reproductive health education, secondary schools, Cameroon

1. Introduction

1.1 Introduction to the Problem

Sexual and reproductive health (SRH) is a fundamental component of overall health and well-being. The World Health Organization defines sexual and reproductive health as a state of complete physical, mental, emotional, and social well-being in all matters relating to sexuality and reproduction rather than merely the absence of disease or infirmity (World Health Organization, 2025, pp. 1–2). Adolescence represents a critical developmental period characterized by rapid biological, psychological, emotional, and social transitions that significantly influence future health behaviors and life outcomes (Sawyer et al., 2018, pp. 223–225). During this period, adolescents often experience increasing curiosity about sexuality while simultaneously facing limited access to accurate and age-appropriate reproductive health information.

1.2 Global Perspective on Adolescent Sexual and Reproductive Health

Globally, adolescent sexual and reproductive health remains a major public health challenge. According to the United Nations Population Fund, millions of adolescents continue to experience unintended pregnancies, sexually transmitted infections (STIs), unsafe abortions, and gender-based violence due to inadequate access to comprehensive sexuality education and reproductive health services (UNFPA, 2024, pp. 8–12). The World Health Organization estimates that complications related to pregnancy and childbirth remain among the leading causes of death among adolescent girls aged 15–19 years worldwide (World Health Organization, 2024, pp. 3–5). Furthermore, adolescents account for a substantial proportion of new STI infections annually, highlighting the urgent need for improved sexual health education and preventive interventions.

1.3 Comprehensive Sexuality Education and Adolescent Health

Comprehensive sexuality education (CSE) has emerged as one of the most effective public health strategies for improving adolescent reproductive health outcomes. UNESCO emphasizes that comprehensive sexuality education equips adolescents with scientifically accurate information, promotes healthy relationships, strengthens decision-making skills, and reduces risky sexual behaviors (UNESCO, 2023, pp. 10–18). Evidence from multiple countries demonstrates that adolescents exposed to comprehensive sexuality education are more likely to delay sexual debut, use contraceptive methods effectively, and seek reproductive health services when needed (Kim et al., 2023, pp. 4–11; Sekhar et al., 2024, pp. 2–7).

1.4 Adolescent Sexual and Reproductive Health in Sub-Saharan Africa

Despite the recognized benefits of comprehensive sexuality education, access remains limited in many low- and middle-income countries. UNICEF reports that adolescents in sub-Saharan Africa continue to face significant barriers to reproductive health information due to cultural taboos,

gender inequalities, inadequate educational resources, and weak health systems (UNICEF, 2024, pp. 12–18). These challenges contribute to high rates of adolescent pregnancy, school dropout, sexually transmitted infections, and poor reproductive health outcomes throughout the region.

In many African societies, socio-cultural norms discourage open discussions about sexuality between parents and adolescents. Numerous studies have demonstrated that inadequate parent-child communication contributes significantly to reproductive health misinformation and risky sexual behaviors among adolescents (Bastien et al., 2011, pp. 8–12; Okyere et al., 2024, pp. 5–8). Adolescents frequently rely on peers, social media platforms, and internet sources for sexual health information, which may expose them to inaccurate information and harmful stereotypes.

1.5 Sexual and Reproductive Health Education in Cameroon

In Cameroon, adolescents constitute a substantial proportion of the national population, making adolescent reproductive health a priority public health concern. National demographic reports indicate persistently high rates of adolescent pregnancy, school dropout, and sexually transmitted infections among young people (National Institute of Statistics Cameroon, 2021, pp. 55–62). Although some reproductive health topics are incorporated into school curricula, studies suggest that adolescents often perceive the information provided as insufficient, fragmented, or inadequately adapted to their practical needs (Foumane et al., 2019, pp. 3–7). Moreover, cultural and religious sensitivities surrounding sexuality frequently limit meaningful discussions within families and educational institutions.

1.6 Research Gap

Recent evidence continues to demonstrate that inadequate sexual and reproductive health education contributes significantly to adverse health and social outcomes among adolescents. Okyere et al. (2024, pp. 6–9) reported that adolescents with poor reproductive health knowledge were significantly more likely to engage in risky sexual behaviors and less likely to utilize reproductive health services. Similarly, Kim et al. (2023, pp. 7–12) found that comprehensive sexuality education substantially improves sexual health knowledge, attitudes, and preventive practices among adolescents.

Despite growing recognition of adolescent reproductive health challenges, limited evidence exists regarding the specific factors associated with inadequate sexual and reproductive health education among secondary school adolescents in Douala, Cameroon. Furthermore, few studies have simultaneously examined both the determinants and consequences of inadequate reproductive health education within this population.

1.7 Hypotheses and Their Correspondence to Research Design

The study was guided by the hypothesis that educational, socio-economic, environmental, cultural, and demographic factors are significantly associated with inadequate sexual and reproductive health education among adolescents in selected secondary schools in Douala, Cameroon. To test this hypothesis, a descriptive cross-sectional quantitative research design was

employed, enabling the assessment of relationships between variables at a single point in time. Chi-square tests were used to determine statistical associations between independent variables and inadequate sexual and reproductive health education, while multivariate logistic regression analysis was performed to identify independent predictors. This design was appropriate for examining the factors associated with inadequate sexual and reproductive health education among adolescents.

2. Methods

2.1 Study Design

A descriptive cross-sectional quantitative study was conducted to assess factors associated with inadequate sexual and reproductive health education among adolescents and to identify the consequences associated with inadequate reproductive health knowledge in selected secondary schools in Douala, Cameroon. The study employed a school-based survey approach and was conducted between February and June 2025.

The cross-sectional design was considered appropriate because it enabled the assessment of multiple educational, socio-economic, cultural, and environmental factors simultaneously within a relatively large adolescent population. Furthermore, this design facilitated the estimation of the prevalence of factors associated with inadequate sexual and reproductive health education and the identification of perceived consequences among participants.

2.2 Study Area

The study was conducted in Douala, the economic capital of Cameroon and the largest city in the country. Douala is characterized by rapid urbanization, a large adolescent population, cultural diversity, and varying socio-economic conditions that influence access to reproductive health information and services.

Three secondary schools were selected for the study were Collège Privé Laïque La Roche, Collège Pascal Tohoua Kamga and Lycée Technique de Douala-Bassa. These schools were selected because they serve adolescents from diverse socio-economic and cultural backgrounds and provide an appropriate setting for investigating factors related to sexual and reproductive health education.

2.3 Study Population

The study population consisted of adolescents enrolled in the selected secondary schools during the study period.

2.3.1 Inclusion Criteria

Participants were eligible for inclusion in the study if they were enrolled in one of the selected secondary schools, were aged between 12 and 19 years, were present during the data collection

period, and provided informed assent with parental or guardian consent to participate in the study.

2.3.2 Exclusion Criteria

Students were excluded from the study if they declined to participate, failed to obtain parental or guardian consent, or submitted incomplete questionnaires that could compromise the quality and completeness of the data collected.

2.4 Sample Size Determination

The minimum sample size was calculated using Cochran's formula for cross-sectional studies:

$$n = Z^2P(1-P)/d^2$$

Where:

- n = minimum required sample size
- Z = 1.96 corresponding to a 95% confidence level
- P = estimated prevalence (50%) due to lack of a precise local estimate
- d = margin of error of 5%

Substituting these values:

$$n = (1.96)^2 \times 0.5 \times (1-0.5)/(0.05)^2$$

$$n = 384$$

Considering possible non-response and incomplete questionnaires, 400 questionnaires were distributed.

2.5 Sampling Technique

A non-probability convenience sampling technique was employed.

Following authorization from school authorities, eligible students who were available during the study period were invited to participate. Participants were recruited consecutively until the required sample size was attained.

Although convenience sampling facilitated access to respondents and enabled efficient data collection, it may have introduced selection bias and may limit the generalizability of the findings beyond the study population.

2.6 Data Collection Instrument

Data were collected using a structured self-administered questionnaire developed after an extensive review of literature on adolescent sexual and reproductive health. The structured questionnaire used for data collection consisted of five sections designed to capture comprehensive information related to adolescents' sexual and reproductive health education. Section A collected socio-demographic information, including age, gender, religion, educational level, school type, and living arrangements. Section B assessed educational factors such as the

availability and quality of sexuality education programs, parent–child communication regarding sexuality, and sources of reproductive health information. Section C explored socio-economic, environmental, and cultural influences, including household economic conditions, access to educational resources, cultural and religious beliefs, peer influence, and media exposure. Section D examined participants' perceptions of the consequences of inadequate sexual education, including sexually transmitted infections, early pregnancies, unsafe abortions, school dropout, and sexual exploitation. Finally, Section E assessed adolescents' educational needs by identifying their preferred reproductive health information and support services.

2.7 Validity and Reliability of the Instrument

Content validity was established through expert review by specialists in public health, adolescent health, reproductive health, and educational research. The questionnaire was pretested among 30 adolescents attending a secondary school not included in the final study sample. Feedback obtained during pilot testing was used to revise unclear items and improve questionnaire structure. Reliability analysis was performed using Cronbach's alpha coefficient. The overall reliability coefficient obtained was 0.82, indicating good internal consistency and reliability of the instrument.

2.8 Data Collection Procedure

Following ethical approval and administrative authorization, trained research assistants visited the selected schools and explained the objectives of the study to eligible participants.

Parental consent and participant assent were obtained prior to questionnaire administration.

Questionnaires were completed anonymously in classroom settings under the supervision of the research team. Participants were encouraged to answer honestly and were assured that all responses would remain confidential.

Completed questionnaires were reviewed daily for completeness and consistency before data entry.

2.9 Data Management and Statistical Analysis

Completed questionnaires were coded and entered into Microsoft Excel before being exported to the Statistical Package for Social Sciences (SPSS) version 25 for analysis. Descriptive statistics were used to summarize and present the study data. These included frequencies and percentages for categorical variables, as well as means and standard deviations for continuous variables. Results were presented using tables and figures. Inferential statistical analyses were conducted to determine associations between selected variables and inadequate sexual and reproductive health education. Chi-square tests were performed to assess associations between categorical variables. Variables with p-values less than 0.05 during bivariate analysis were entered into a multivariate logistic regression model to identify independent predictors of inadequate sexual and

reproductive health education. Adjusted Odds Ratios (AORs), 95% Confidence Intervals (95% CI), and p-values were reported. Statistical significance was established at $p < 0.05$.

2.10 Ethical Considerations

Ethical approval for this study was obtained from the University of Douala Ethics Committee under approval number 3476 IEC-UD/06/2025/T. Administrative authorization was subsequently obtained from the administrations of the selected secondary schools before commencement of data collection. Because study participants were minors, written informed consent was obtained from parents or legal guardians, while written assent was obtained from participating adolescents. Participation was voluntary, and respondents were informed of their right to withdraw from the study at any stage without any consequences. Confidentiality and anonymity were strictly maintained throughout the study. No personal identifiers were collected, and all questionnaires were coded using unique identification numbers. Data were stored securely and accessed only by members of the research team.

3. Results

3.1 Socio-Demographic Characteristics of Respondents

Out of the 400 questionnaires distributed, a total of 378 correctly completed questionnaires were returned and retained for analysis, representing a response rate of 94.5%. Table 1 presents the socio-demographic characteristics of the 378 adolescents who participated in the study. The majority of respondents were female (74.1%), while males represented 25.9% of the study population. Nearly half of the participants (48.1%) were aged between 15 and 16 years. Most respondents attended private secular schools (70.4%), identified as Catholics (57.4%), and lived with both parents (52.6%).

Table 1: Socio-demographic characteristics of respondents (n = 378)

Variable	Number of Respondents (n)	Percentage (%)
Gender		
Female	280	74.1
Male	98	25.9
Age Group (Years)		
12–14	116	30.7
15–16	182	48.1
17–19	80	21.2
Type of School		
Private secular	266	70.4
Public	112	29.6
Religion		
Catholic	217	57.4
Protestant	93	24.6
Muslim	42	11.1
Others	26	6.9
Living Arrangement		
Both parents	199	52.6
Single parent	128	33.9
Guardian/Relatives	51	13.5

3.2 Educational Factors Associated with Inadequate Sexual and Reproductive Health Education

Table 2 presents the educational factors associated with inadequate sexual and reproductive health education among adolescents. More than half of the respondents (57.8%) reported the absence of structured sexual and reproductive health education programs in schools. Furthermore, 56.6% indicated that they rarely or never discussed sexual matters with their parents, while 92.1% believed that comprehensive sexuality education was insufficient within their schools.

Table 2: Educational factors associated with inadequate sexual and reproductive health education

Educational Factors	Number of Respondents (n)	Percentage (%)	95% CI
Absence of SRH programs in schools	219	57.84	52.8–62.8
Lack of parent-child communication	214	56.61	51.6–61.5
Inadequate comprehensive sexuality education	348	92.06	89.3–94.8
Lack of trained sexuality educators	173	45.77	40.8–50.8
Lack of educational materials	165	43.65	38.7–48.6

3.3 Socio-Economic, Environmental, and Cultural Factors

Table 3 summarizes the socio-economic, environmental, and cultural factors perceived to influence sexual and reproductive health education among adolescents. Nearly half of respondents (47.4%) identified inadequate financial resources as an important barrier. Media and social media influence was reported by 36.3% of participants, while cultural and religious restrictions were identified by 24.3%.

Table 3: Socio-Economic, Environmental, and Cultural Factors Associated with Inadequate Sexual and Reproductive Health Education

Factor	Number of Respondents (n)	Percentage (%)	95% CI
Lack of financial resources	179	47.35	42.3–52.4
Media/social media influence	137	36.25	31.4–41.1
Cultural/religious restrictions	92	24.34	20.0–28.7
Peer influence	128	33.86	29.1–38.6
Limited access to health information	154	40.74	35.8–45.7

3.4 Consequences of Inadequate Sexual and Reproductive Health Education

Figure 1 presents the major consequences associated with inadequate sexual and reproductive health education among adolescents. Sexually transmitted infections were the most frequently reported consequence (62.5%), followed by early and unintended pregnancies (53.4%). Other notable consequences included school dropout (38.0%), unsafe abortions (29.0%), and sexual exploitation (26.0%).

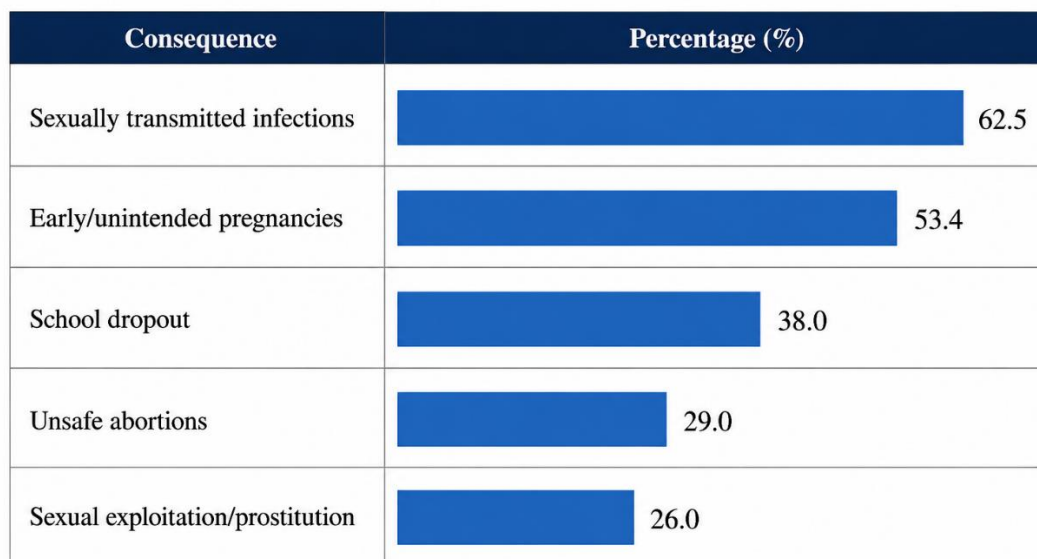


Figure 1: Major consequences of inadequate sexual and reproductive health education among adolescents

3.5 Adolescents' Needs Regarding Sexual and Reproductive Health Education

Table 4 presents the educational needs expressed by respondents. The majority of adolescents (70.6%) indicated the need for guidance from teachers, parents, and healthcare professionals. More than half (56.9%) desired regular classroom discussions on reproductive health issues.

Table 4: Adolescents' needs regarding sexual and reproductive health education

Educational Needs	Number of Respondents (n)	Percentage (%)
Counseling and orientation	176	46.56
Information on contraception and STIs	116	30.69
Regular classroom discussions	215	56.88
Guidance from teachers, parents, and health workers	267	70.63
Youth-friendly reproductive health services	194	51.32

3.6 Factors Associated with Inadequate Sexual and Reproductive Health Education

Table 5 presents the results of the Chi-square analysis. Significant associations were observed between inadequate sexual and reproductive health education and several variables including gender, type of school, parent-child communication, availability of school-based SRH programs, media influence, and cultural or religious restrictions. The strongest associations were observed for the absence of school-based sexual and reproductive health programs and poor parent-child communication regarding sexuality.

Table 5: Bivariate analysis of factors associated with inadequate sexual and reproductive health education

Variable	χ^2	p-value
Gender	5.82	0.016
Age group	4.17	0.041
Type of school	8.94	0.003
Parent-child communication	18.76	<0.001
Availability of SRH programs	22.43	<0.001
Media/social media influence	7.15	0.008
Cultural/religious restrictions	6.37	0.012

3.7 Independent Predictors of Inadequate Sexual and Reproductive Health Education

Variables found significant during bivariate analysis were entered into a multivariate logistic regression model. Table 6 presents the independent predictors of inadequate sexual and reproductive health education among adolescents. Multivariate analysis revealed that the absence of school-based sexual and reproductive health programs was the strongest independent predictor of inadequate reproductive health education. Adolescents who reported poor parent-child

communication were approximately three times more likely to experience inadequate reproductive health education compared to those who regularly discussed sexuality with their parents

Table 6: Multivariate logistic regression analysis of predictors of inadequate sexual and reproductive health education

Variable	AOR	95% CI	p-value
Absence of SRH programs in schools	3.48	2.01–6.04	<0.001
Lack of parent-child communication	2.96	1.71–5.13	<0.001
Media/social media influence	1.89	1.08–3.31	0.024
Cultural/religious restrictions	1.76	1.02–3.05	0.041
Female gender	1.58	0.92–2.73	0.093

3.8 Gender-Based Analysis of Factors and Consequences

Table 7 compares selected factors and consequences according to gender. Female adolescents more frequently reported concerns regarding early and unintended pregnancies, whereas male adolescents more frequently identified media and peer influence as important contributors to risky sexual behaviors. The findings indicate that although inadequate sexual and reproductive health education affects both genders, the specific concerns and educational needs differ significantly between male and female adolescents.

Table 7: Gender differences in factors and consequences of inadequate sexual and reproductive health education.

Variable	Female (%)	Male (%)	p-value
Early/unintended pregnancy concerns	61.4	31.6	<0.001
Media/social media influence	33.2	44.9	0.028
Need for counseling and guidance	74.3	59.2	0.011
Lack of parent-child communication	59.6	48.0	0.034

4. Discussion

The present study investigated factors associated with inadequate sexual and reproductive health (SRH) education among adolescents and examined the consequences of inadequate SRH knowledge in selected secondary schools in Douala, Cameroon. The findings revealed important educational, socio-economic, environmental, and cultural barriers that continue to limit adolescents' access to accurate reproductive health information. Furthermore, the study demonstrated that inadequate SRH education is associated with several adverse health and social outcomes, including sexually transmitted infections, unintended pregnancies, school dropout, unsafe abortions, and sexual exploitation.

The socio-demographic profile of respondents showed a predominance of female participants (74.1%), with nearly half of respondents aged between 15 and 16 years. This age group

represents mid-adolescence, a developmental stage characterized by increasing sexual curiosity, experimentation, and vulnerability to reproductive health risks (Sawyer et al., 2018). Similar age distributions have been reported in studies conducted among secondary school adolescents in Cameroon, Ghana, Kenya, and Nigeria, where adolescents aged 15–17 years constitute the group most exposed to reproductive health challenges (Okyere et al., 2024; UNICEF, 2024).

One of the major findings of this study was the absence of structured school-based sexual and reproductive health programs, identified by 57.8% of respondents as an important contributor to inadequate SRH education. The multivariate analysis further revealed that adolescents attending schools without structured SRH programs were approximately three and a half times more likely to experience inadequate reproductive health education compared to their counterparts (AOR = 3.48; 95% CI: 2.01–6.04; $p < 0.001$). This finding is consistent with UNESCO (2023), which reported that comprehensive sexuality education significantly improves adolescents' reproductive health knowledge, attitudes, and preventive behaviors. Similar observations have been reported in several African countries where inadequate implementation of comprehensive sexuality education has been associated with poor reproductive health literacy and increased sexual risk-taking among adolescents (Kim et al., 2023; Sekhar et al., 2024).

The study also identified poor parent-child communication regarding sexuality as a major determinant of inadequate SRH education. More than half of respondents reported rarely or never discussing sexual matters with their parents. Logistic regression analysis demonstrated that adolescents lacking parental communication were nearly three times more likely to experience inadequate reproductive health education (AOR = 2.96; 95% CI: 1.71–5.13; $p < 0.001$). These findings support previous studies conducted in sub-Saharan Africa, which have consistently highlighted the protective role of open parent-child communication in reducing risky sexual behaviors and improving reproductive health outcomes (Bastien et al., 2011; Okyere et al., 2024). Cultural taboos, religious beliefs, embarrassment, and limited parental knowledge may explain the persistence of communication barriers within many families.

Socio-economic factors also emerged as important determinants of inadequate SRH education. Nearly half of respondents identified inadequate financial resources as a significant barrier. Resource limitations may reduce schools' ability to implement sexuality education programs, train educators, and provide educational materials. Similar findings have been reported by UNFPA (2024), which noted that inadequate investment in adolescent reproductive health programs continues to undermine educational interventions across many low- and middle-income countries.

Media and social media influence constituted another important factor associated with inadequate reproductive health education. Adolescents reporting substantial media influence were significantly more likely to exhibit inadequate SRH knowledge (AOR = 1.89; 95% CI: 1.08–3.31; $p = 0.024$). The rapid expansion of internet access and social networking platforms has transformed the way adolescents obtain sexual information. Although digital media can facilitate access to reproductive health information, it may also expose young people to

misinformation, sexually explicit content, unrealistic perceptions of sexuality, and harmful social pressures. Similar concerns have been reported by UNICEF (2024) and WHO (2024), both of which emphasize the importance of media literacy as a component of adolescent sexuality education programs.

Cultural and religious restrictions were also significantly associated with inadequate SRH education. Adolescents reporting cultural or religious barriers were significantly more likely to have inadequate reproductive health knowledge (AOR = 1.76; 95% CI: 1.02–3.05; $p = 0.041$). These findings are consistent with previous studies showing that conservative cultural norms frequently discourage open discussions about sexuality and limit adolescents' access to scientifically accurate reproductive health information (UNESCO, 2023; UNFPA, 2024).

The present study identified sexually transmitted infections (62.5%) as the most frequently reported consequence of inadequate SRH education, followed by early and unintended pregnancies (53.4%). These findings corroborate global evidence demonstrating that inadequate reproductive health knowledge increases vulnerability to unsafe sexual practices, inconsistent contraceptive use, and delayed healthcare seeking (WHO, 2024). The high prevalence of reported sexually transmitted infections and unintended pregnancies observed in this study underscores the urgent need for comprehensive interventions targeting adolescent reproductive health.

School dropout emerged as another important consequence identified by respondents. Adolescent pregnancies frequently interrupt educational trajectories, reducing opportunities for academic achievement and future socio-economic advancement. Similar findings have been documented throughout sub-Saharan Africa, where adolescent pregnancy remains a major contributor to school discontinuation among girls (UNICEF, 2024; UNFPA, 2024).

Gender-stratified analysis revealed important differences between male and female adolescents. Female participants were significantly more concerned about unintended pregnancies and expressed greater need for counseling and reproductive health guidance, whereas males more frequently identified media influence as a determinant of risky sexual behavior. These findings suggest that reproductive health interventions should be tailored to address gender-specific needs and vulnerabilities among adolescents.

The findings of this study can be interpreted within the framework of Social Learning Theory. According to this theory, adolescents acquire behaviors, attitudes, and beliefs through observation and interaction with influential individuals and social environments. In the present study, inadequate parental communication, limited school-based education, peer influence, and media exposure collectively contributed to reproductive health knowledge gaps among adolescents. Strengthening positive educational influences within schools, families, and communities may therefore enhance adolescents' ability to make informed reproductive health decisions.

Overall, the study demonstrates that inadequate sexual and reproductive health education remains a significant public health challenge among adolescents in Douala. Addressing educational deficiencies, strengthening parent-child communication, improving access to comprehensive sexuality education, and promoting adolescent-friendly reproductive health services are likely to contribute substantially to improved adolescent reproductive health outcomes.

Despite its strengths, the study has several limitations. First, the cross-sectional design limits the ability to establish causal relationships between identified factors and inadequate sexual and reproductive health education. Second, the use of a non-probability convenience sampling technique may have introduced selection bias and may limit the generalizability of findings to all adolescents in Douala and Cameroon. Third, the predominance of female respondents may have influenced the observed patterns and reduced opportunities for balanced gender comparisons. Fourth, information was self-reported and may therefore be subject to recall bias and social desirability bias. Finally, some variables related to sexual behavior and reproductive health outcomes may have been underreported due to the sensitive nature of the subject matter.

This study has demonstrated that inadequate sexual and reproductive health education remains a significant challenge among adolescents attending secondary schools in Douala, Cameroon. The principal factors associated with inadequate reproductive health education included the absence of structured school-based sexuality education programs, poor parent-child communication regarding sexuality, media and social media influence, financial constraints, and cultural or religious restrictions. The most frequently reported consequences were sexually transmitted infections, unintended pregnancies, school dropout, unsafe abortions, and sexual exploitation. Multivariate analysis identified the absence of school-based sexual and reproductive health programs and poor parent-child communication as the strongest independent predictors of inadequate reproductive health education.

Strengthening comprehensive sexuality education, promoting effective parent-adolescent communication, enhancing media literacy, and improving access to adolescent-friendly reproductive health services may substantially improve adolescent reproductive health outcomes and contribute to the reduction of preventable reproductive health problems among young people.

Based on the findings of this study, a comprehensive and collaborative approach is required to improve sexual and reproductive health education among adolescents. Schools should integrate comprehensive sexuality education into their curricula, strengthen counseling services, and regularly organize reproductive health awareness activities. Parents and guardians should promote open, age-appropriate communication regarding sexuality and actively participate in adolescent reproductive health education while fostering supportive family environments. Healthcare providers should expand adolescent-friendly reproductive health services, implement school-based outreach programs, and collaborate closely with educational institutions to improve access to accurate reproductive health information. Policymakers should strengthen policies

supporting comprehensive sexuality education, increase funding for adolescent reproductive health programs, and encourage multisectoral partnerships involving the education, health, community, and religious sectors. Additionally, further research is needed to explore the long-term effects of reproductive health education, investigate gender-specific determinants of adolescent reproductive health knowledge and behaviors, and evaluate the effectiveness of school-based comprehensive sexuality education interventions in Cameroon. Collectively, these measures have the potential to improve adolescents' reproductive health knowledge, promote healthy behaviors, and reduce the burden of sexually transmitted infections, unintended pregnancies, school dropout, and other adverse reproductive health outcomes.

Acknowledgements

The authors sincerely thank all adolescents who participated in this study for their valuable contributions. We are equally grateful to the administrations of Collège Privé Laïque La Roche, Collège Pascal Tohoua Kanga, and Lycée Technique de Douala-Bassa for granting permission and facilitating data collection. Special appreciation is extended to the research assistants who contributed to data collection and quality control throughout the study. The authors also acknowledge the support provided by colleagues and mentors who contributed valuable technical and academic guidance during the implementation of this research.

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